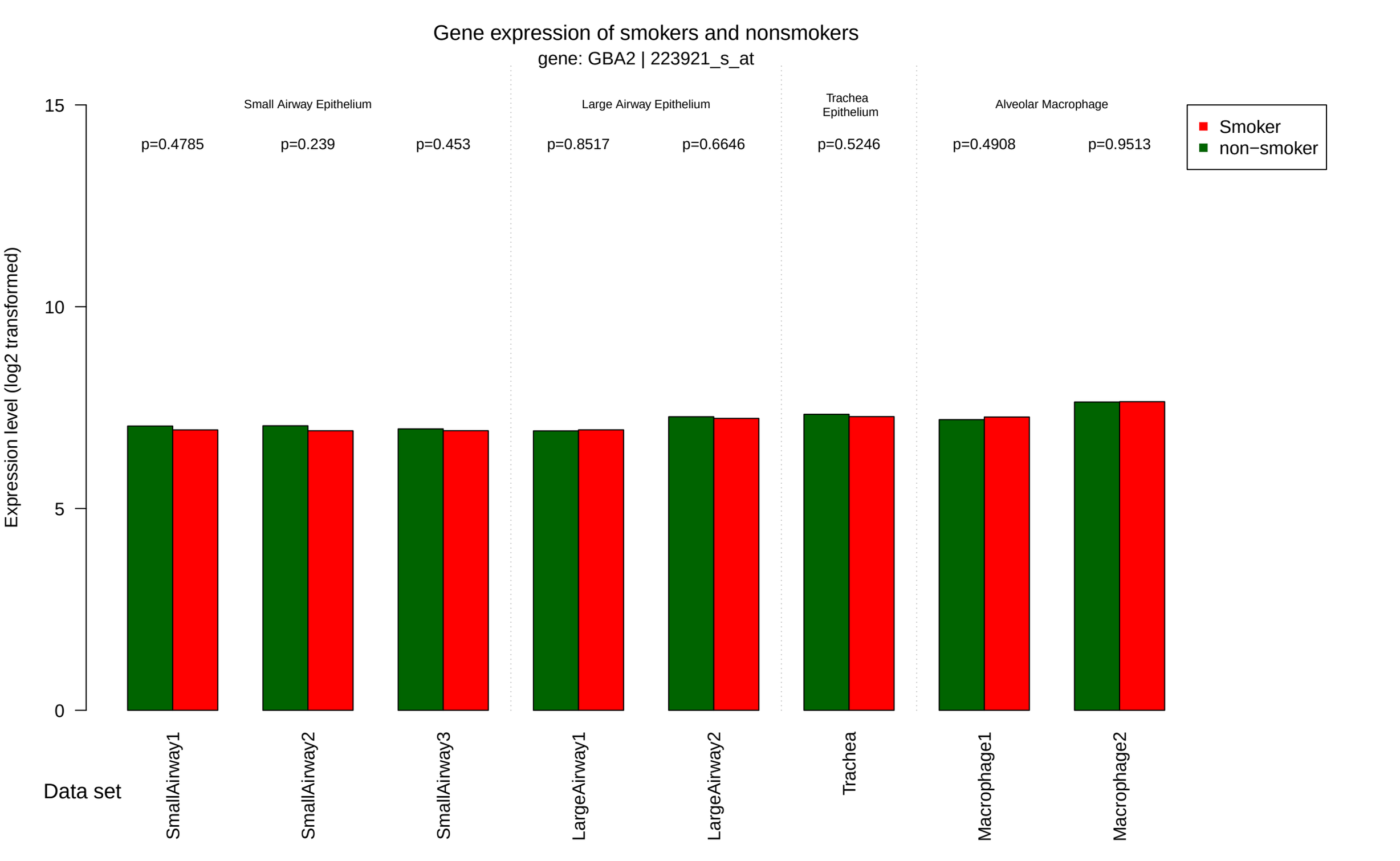
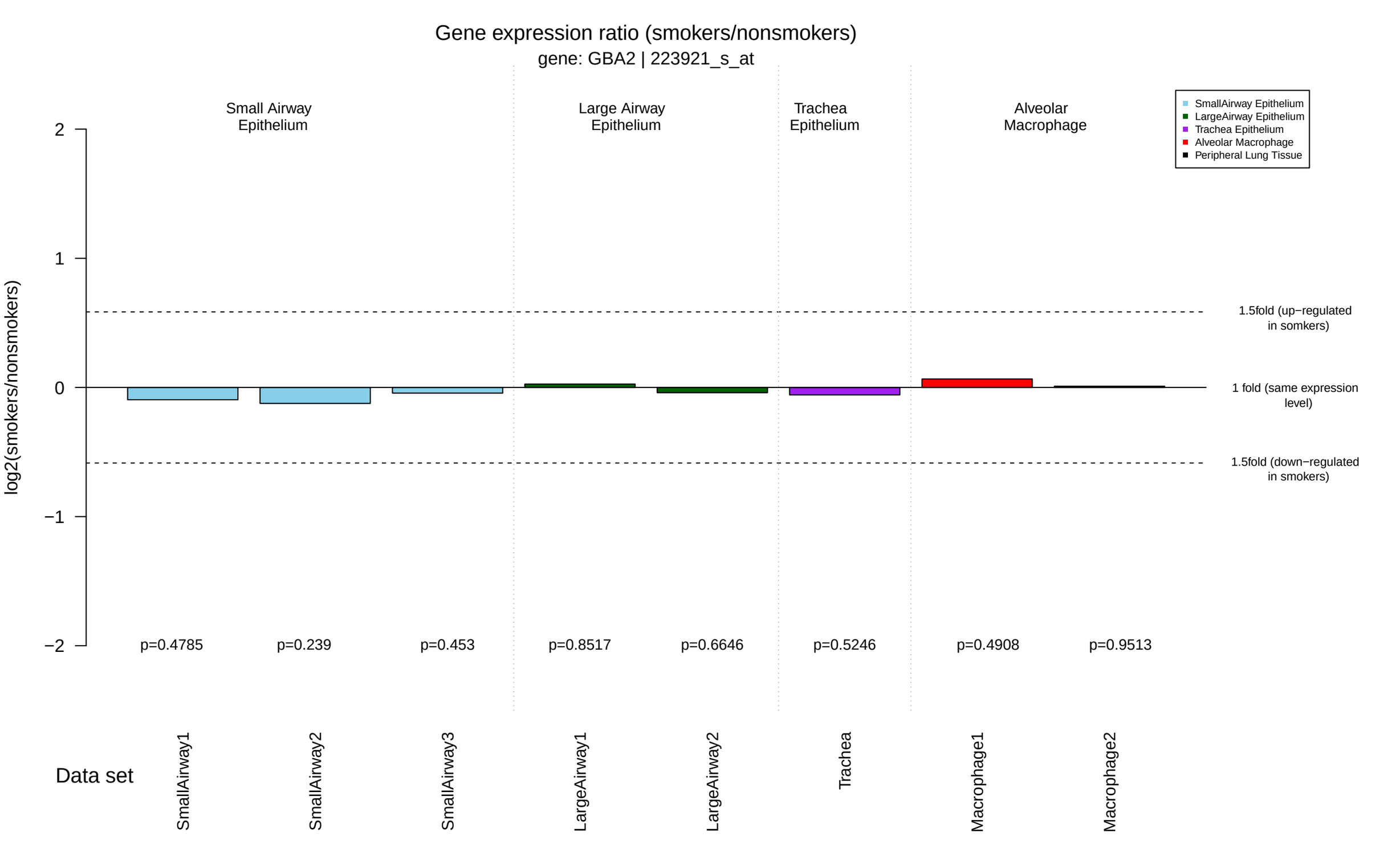
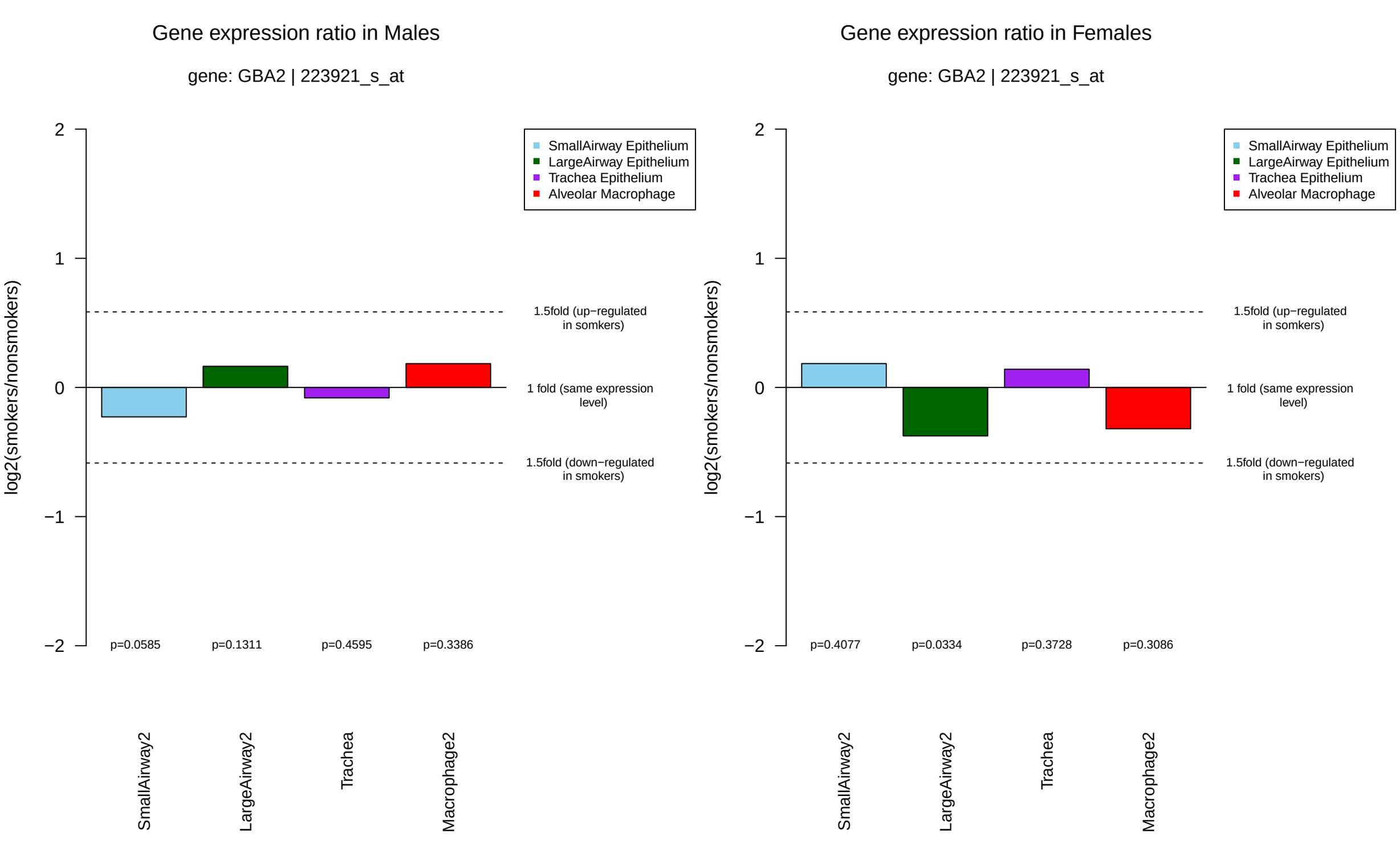
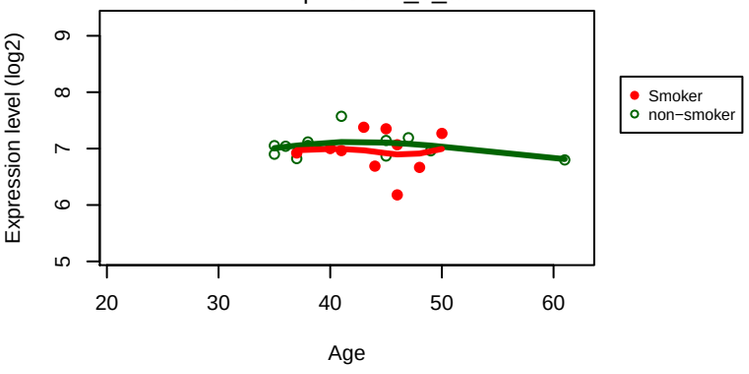
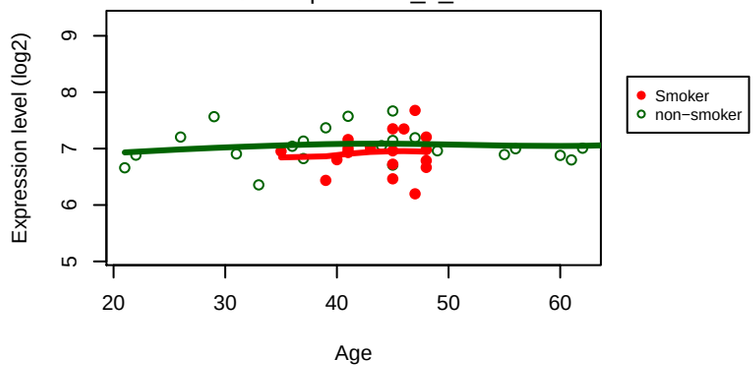
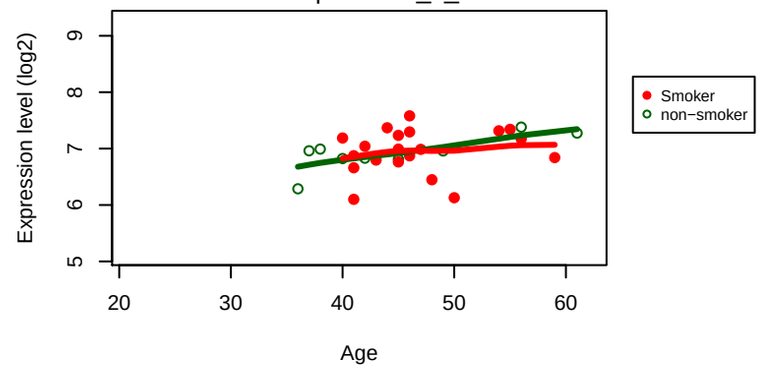

Gene expression of smokers and nonsmokers
gene: GBA2 | 223921_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4785
p=0.239
p=0.453
p=0.8517
p=0.6646
p=0.5246
p=0.4908
p=0.9513
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

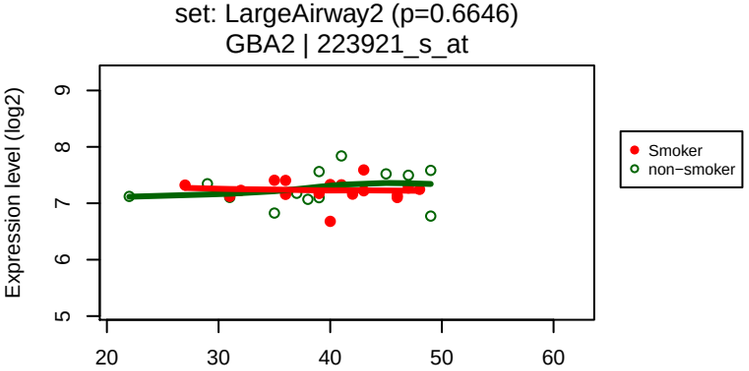

Gene expression ratio (smokers/nonsmokers)
gene: GBA2 | 223921_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4785
p=0.239
p=0.453
p=0.8517
p=0.6646
p=0.5246
p=0.4908
p=0.9513
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

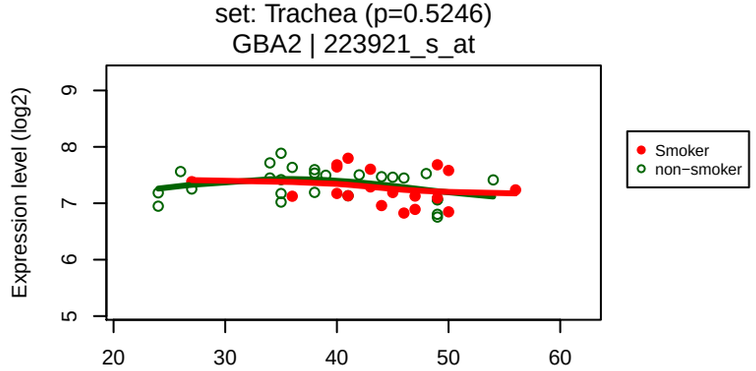

Gene expression ratio in Males
gene: GBA2 | 223921_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0585
p=0.1311
p=0.4595
p=0.3386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GBA2 | 223921_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4077
p=0.0334
p=0.3728
p=0.3086
SmallAirway2
LargeAirway2
Trachea
Macrophage2

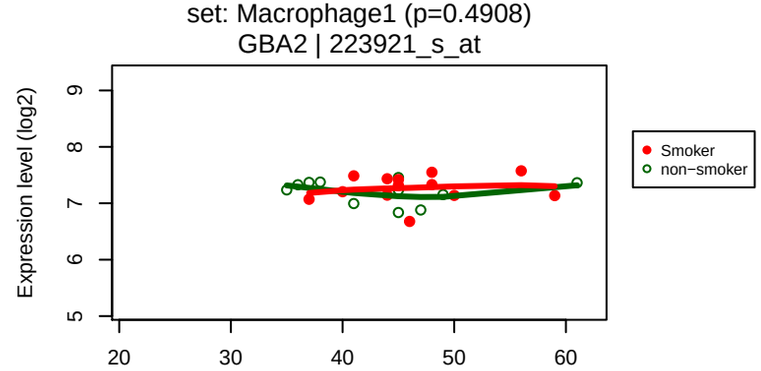

set: SmallAirway1 (p=0.4785)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker

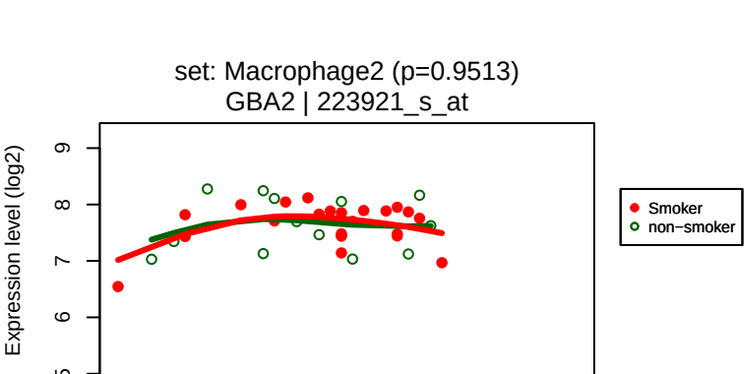

set: SmallAirway2 (p=0.239)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker

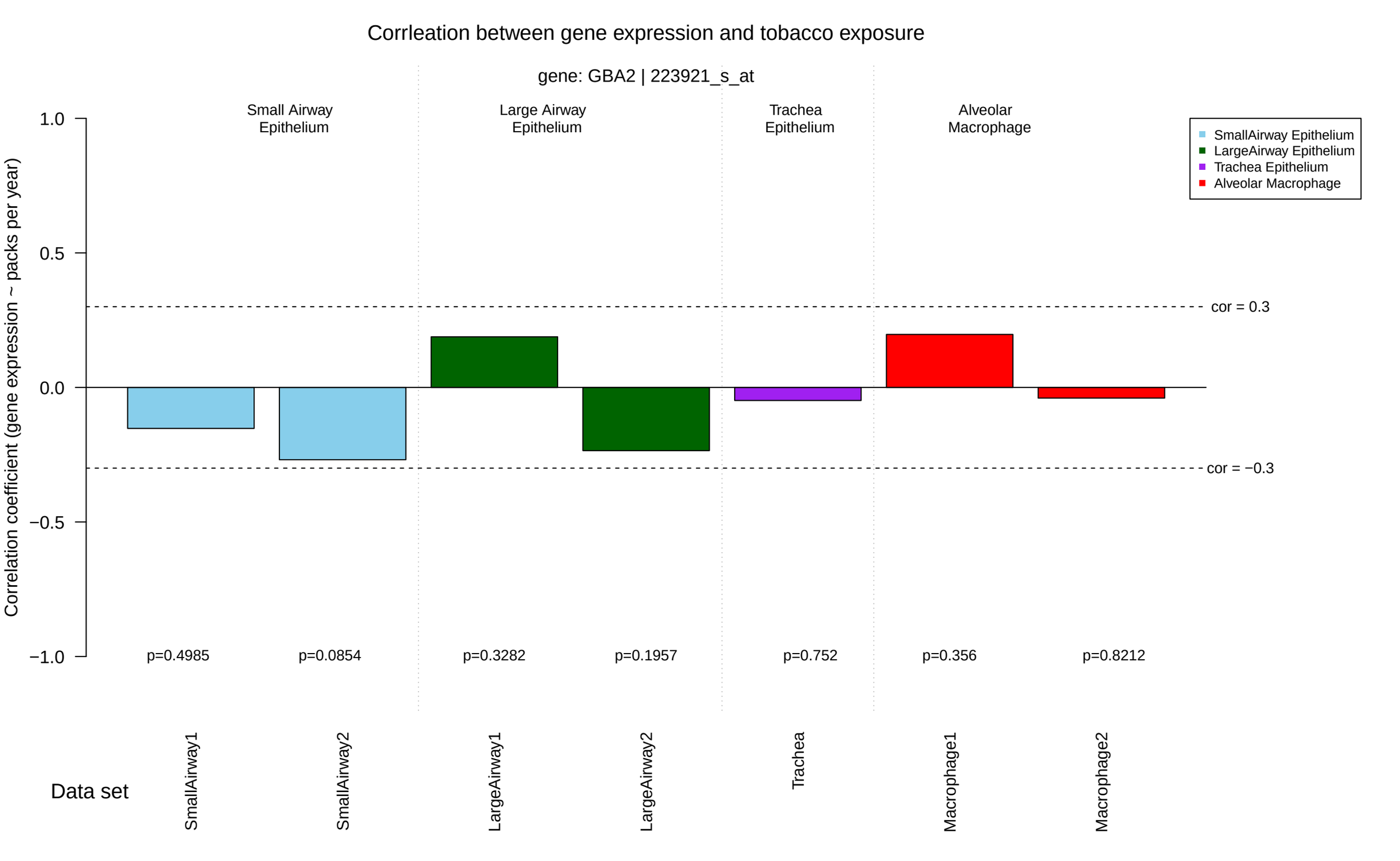

set: LargeAirway1 (p=0.8517)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6646)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5246)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4908)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9513)
GBA2 | 223921_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GBA2 | 223921_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4985
p=0.0854
p=0.3282
p=0.1957
p=0.752
p=0.356
p=0.8212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2